BinFinder MVP — Response to Feedback

Team Alpha · LASU Epe Campus · NEX Research and Innovation Cohort

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Live app	https://waste-management-system-bay.vercel.app/
Source code	https://github.com/OLADELE-OLAWALE/Waste-Management-System

The home screen is no longer a map with three tabs. Opening the link now does one thing: it finds the student, highlights the nearest bin they can actually use, shows how far it is and draws the route — before anything is tapped. Every other change in this report supports that one action.

The screens in sections 5 and 6 are the live app, captured on a phone-sized screen, not mock-ups.

1. The six fixes

#	What was asked	What we changed	Status
1	Make “Find nearest bin” the main home action. GPS automatically, nearest available bin highlighted with its distance and route. Remove the Nearest tab.	The tab bar is gone. On open the app asks for location, picks the nearest bin with status Available, and shows one card: distance in metres, bin name, an Available chip, walking time, and when the status was last updated. The route is drawn at the same moment. If a closer bin was skipped, the card says which and why. No tap is needed to reach any of this.	Done
2	Redesign the map pins: status by colour and shape/icon, larger pins, cluster overlapping pins.	Pins are 34 px and carry status three ways at once — green circle with ✓ = available, red diamond with ! = overflowing, grey square with ✕ = damaged. Solid fill with a white border so they hold up in sunlight and against satellite imagery. Bins that would overlap collapse into a neutral dark cluster showing the count; below zoom 18 they separate again.	Done
3	Simplify the bottom panel: main action plus a collapsible legend. Remove “Location set from map” and the unused “You” legend entry.	The panel is now the answer card, one primary button and one small text link. “Location set from map”, “16 bins on campus”, the GPS button and the “You” legend entry are all gone. The legend is folded behind “What the pins mean” and holds three entries.	Done
4	Separate the two reporting flows: “Report a bin issue” and “Request a bin here”, appearing separately in the admin dashboard.	“Report a problem” opens a two-way choice. Bin issue: pick the bin, tap Full/overflowing or Damaged. Request a bin: drag a pin to the spot, pick one of four reasons. The dashboard's Reports tab now has two streams with their own counts, and the KPI row counts open bin issues and open bin requests separately.	Done
5	Improve the map: light street map with building labels, satellite still available.	The street map is now the default and satellite is one tap away in the layer control. OpenStreetMap has almost nothing mapped inside LASU Epe, so the campus place names are drawn from our own zone list on top of the tiles; they appear at walking zoom, where they are useful, and stay hidden when the map is pulled back.	Done
6	Clean up the bin popup, add a reporting limit and a confirmation message.	The popup is name, status chip, distance, last-updated time, a Directions button and a small “Report this bin” link — about a fifth of a phone screen. The server now allows one report per device, per bin, per day (per spot within 25 m for bin requests), on top of the existing 30-per-hour network cap, and a single report no longer changes a	Done

#	What was asked	What we changed	Status
		bin's status on its own (section 6). Every report ends on a confirmation screen.	

2. The target MVP flow, step by step

Checked against the five steps as they are written in the feedback.

Step	What the app does now
Open — nearest bin appears with name, distance and availability status; no tab selection	The card carries the bin name, the distance in metres and an "Available" status chip, with the walking time and when the status was last updated underneath. There are no tabs.
Get directions — the student taps the bin, the map shows the route, there is a clear Directions button	Tapping any bin draws the route to that bin straight away and holds it there while the popup is open, including across the 15-second data refresh. The popup's Directions button hands the walk to the phone's own map app, which knows the footpaths.
Report a problem — two options: report a bin issue, or request a bin here	Two separate flows from the first tap. Bin issues name a bin and a fault; bin requests carry a dropped pin and a reason.
Confirm the report — confirmation, and every bin shows a Last updated time	A confirmation screen after every report, with the reference number, what was reported, the bin or spot, what the bin now shows, and how many reports that area has had in 7 days. "Updated 2 d ago" appears on the home card and in every bin popup, not only after reporting.
Admin dashboard — repeated bin problems, repeated "no bin here" requests, bins that are frequently reported, where bins are needed	Zone hotspot ranking with "6 flagged this week" per zone, the two report streams, a "Most reported bins · last 14 days" list above the bin-issue stream (click a row to jump to that bin), and the Allocate tab for where bins should be added.

The five-second target

On a laptop the app takes about 0.2 seconds from opening the link to the answer being on screen. That is not the real test: on a phone the deciding factor is the GPS fix, not our code. The app now accepts a fix the phone already took in the last minute — the usual case if the student has used Maps or WhatsApp recently — and shows a "Finding your location" state immediately rather than a blank screen. We have not yet timed it on campus Wi-Fi, and we will report that figure once we can test on site.

3. The other points in the feedback

- **Feedback after a report (point 5).** Every report now ends on a confirmation screen showing the reference number, what was reported, which bin or spot, what the bin now shows and when it was updated, plus how many reports that area has had in the last 7 days. Previously it was a toast that vanished.
- **"Last updated" (point 5, step 4).** Bins carry a status timestamp. A student report changes the bin's status immediately, so "Updated 2 h ago" is a real time, not a guess.
- **Student and admin sides connected (point 6).** The hotspot table carries a line per zone — "6 flagged this week" — counting distinct devices over the last 7 days, and the bin-request stream shows how far the nearest bin was from each request. The student side and the admin side read as one system.
- **What the feedback said to keep.** Still no accounts, still one link that works from WhatsApp, still Available / Overflowing / Damaged. The reporting limit is keyed to an anonymous per-browser id, not to a login.

4. Wrong reports

The feedback asked for protection against one person distorting the data. A reporting limit answers only part of that, so we went further: a single report no longer decides what everybody else sees. Three rules do the work, and none of them needs an account.

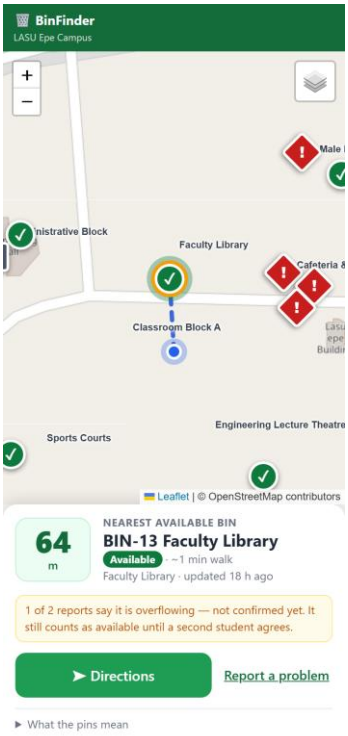
Rule	What it stops
One report per device, per bin, per day — per spot within 25 m for bin requests, which have no bin to key on.	One phone repeating itself, by accident or on purpose.
A bin changes status only when two different phones report the same thing within six hours.	One mistaken or joking report hiding a working bin from everyone.
A status nobody repeats goes stale: overflowing clears itself after 24 hours, damaged after seven days.	A wrong or out-of-date status lasting forever because nobody corrected it.
A cap of 30 reports an hour from one network.	A public link being flooded during a demo or a prank.
The collection crew's "Mark emptied" in the dashboard overrides all of it at once.	The system arguing with someone who is standing at the bin.

Until that second report arrives the bin is flagged, not hidden. It keeps its Available status and still counts as the nearest usable bin, but it wears an amber ring on the map, and both the home card and the popup say "1 of 2 reports say it is overflowing — not confirmed yet". The student who reported it is told on the confirmation screen that the report is counted and what has to happen next, so the feedback they get is honest rather than flattering.

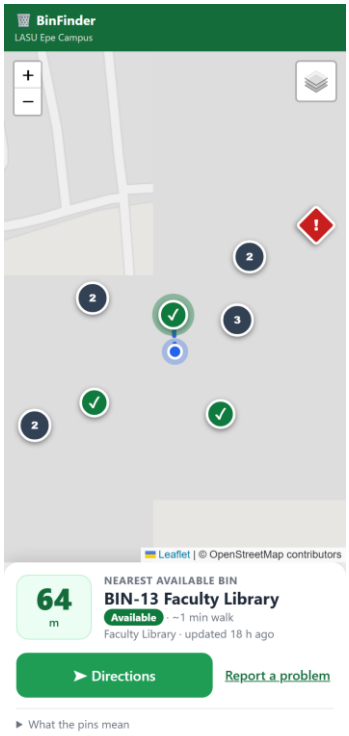
Expiry leaves the timestamp at the moment the information went stale instead of moving it to now, so the app says "updated 1 d ago" rather than pretending someone has just checked. Both thresholds and both expiry windows are editable in the dashboard's Model tab, so the team can tune them once there is real reporting volume.

What we deliberately did not do: require a matric number or a phone number. It would break the no-sign-up property the feedback told us to keep, and it does not actually stop anyone who is willing to use a second phone.

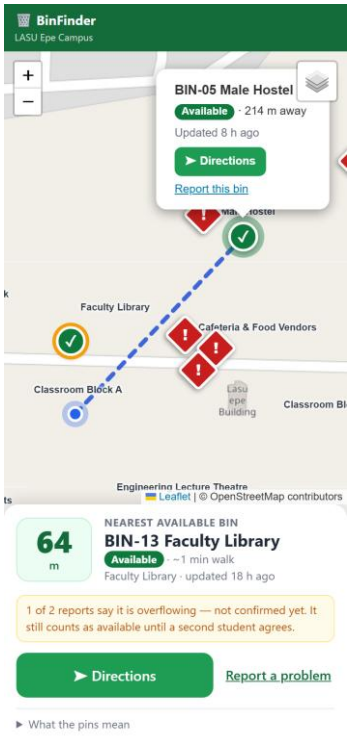
5. The updated student screens



1. Home, nothing tapped: nearest available bin, 64 m, Available, route drawn. (Fix 1)

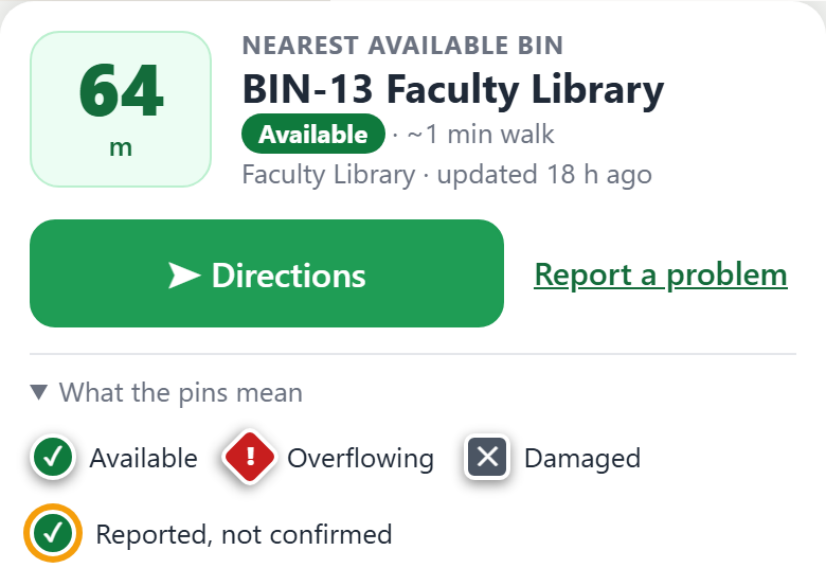


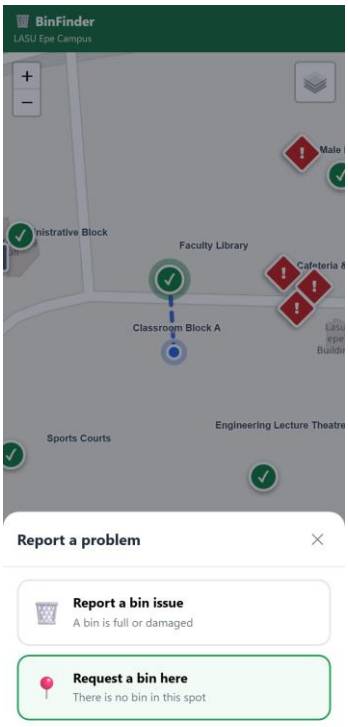
2. Pins by shape and colour; overlapping bins cluster when pulled back. (Fix 2)



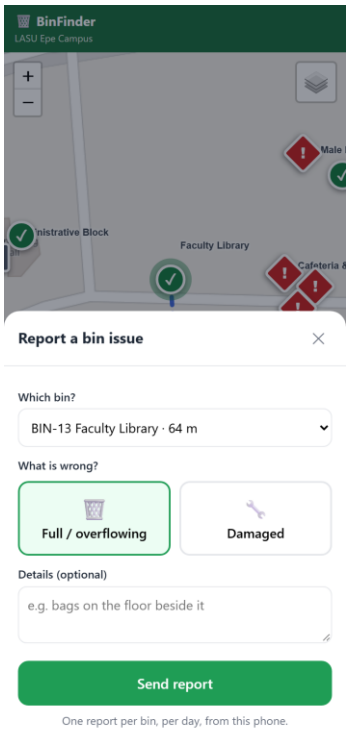
4. Tapping a bin routes to it; the popup shows status, distance, last updated. (Fixes 2, 6)

3. The bottom panel: the answer, one button, one link, legend folded away. (Fix 3)

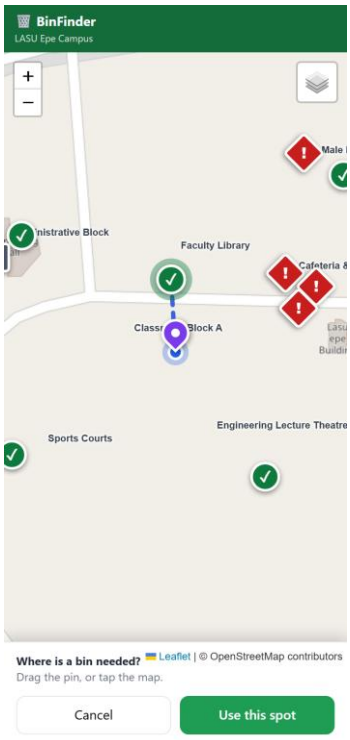




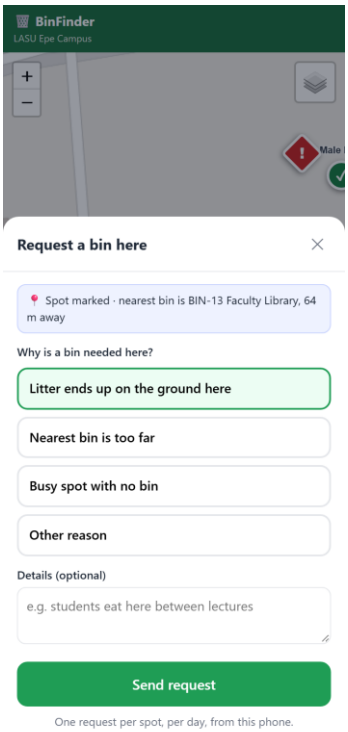
5. Report a problem: two flows, split at the first tap. (Fix 4)



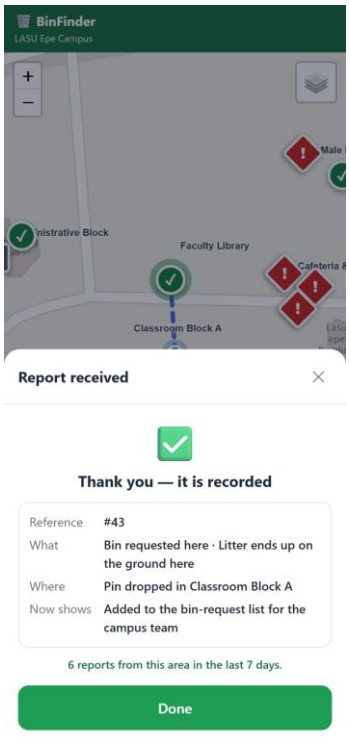
6. Report a bin issue: bin, fault, optional detail. (Fix 4)



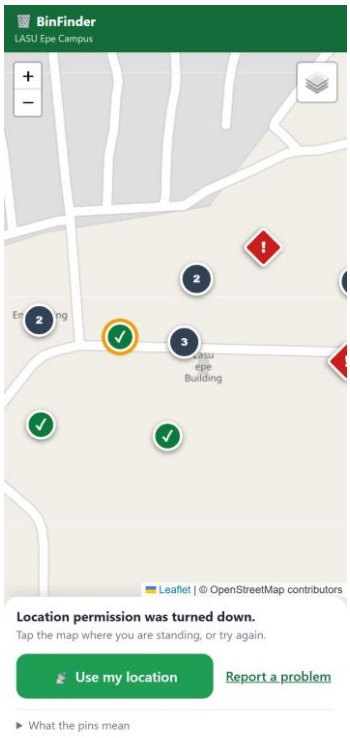
7. Request a bin here: the student drags the pin. (Fix 4)



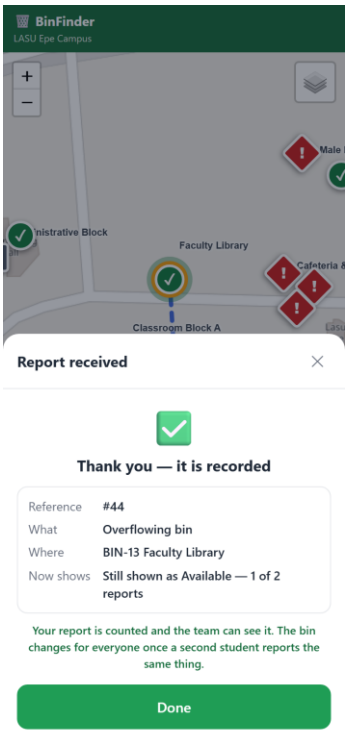
8. Request a bin here: why a bin is needed. (Fix 4)



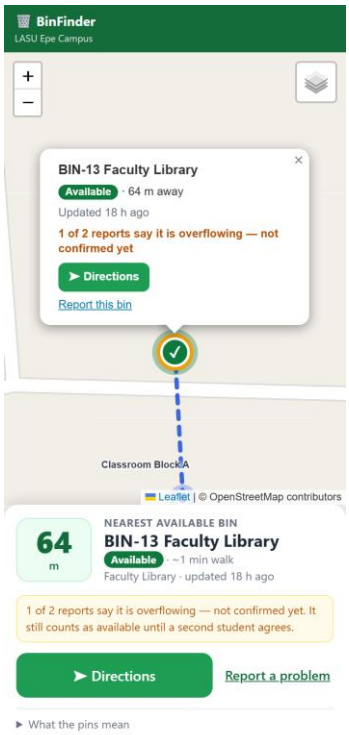
9. Confirmation: reference, what, where, what the bin shows. (Point 5)



10. Location turned down: the app still works by tapping the map.



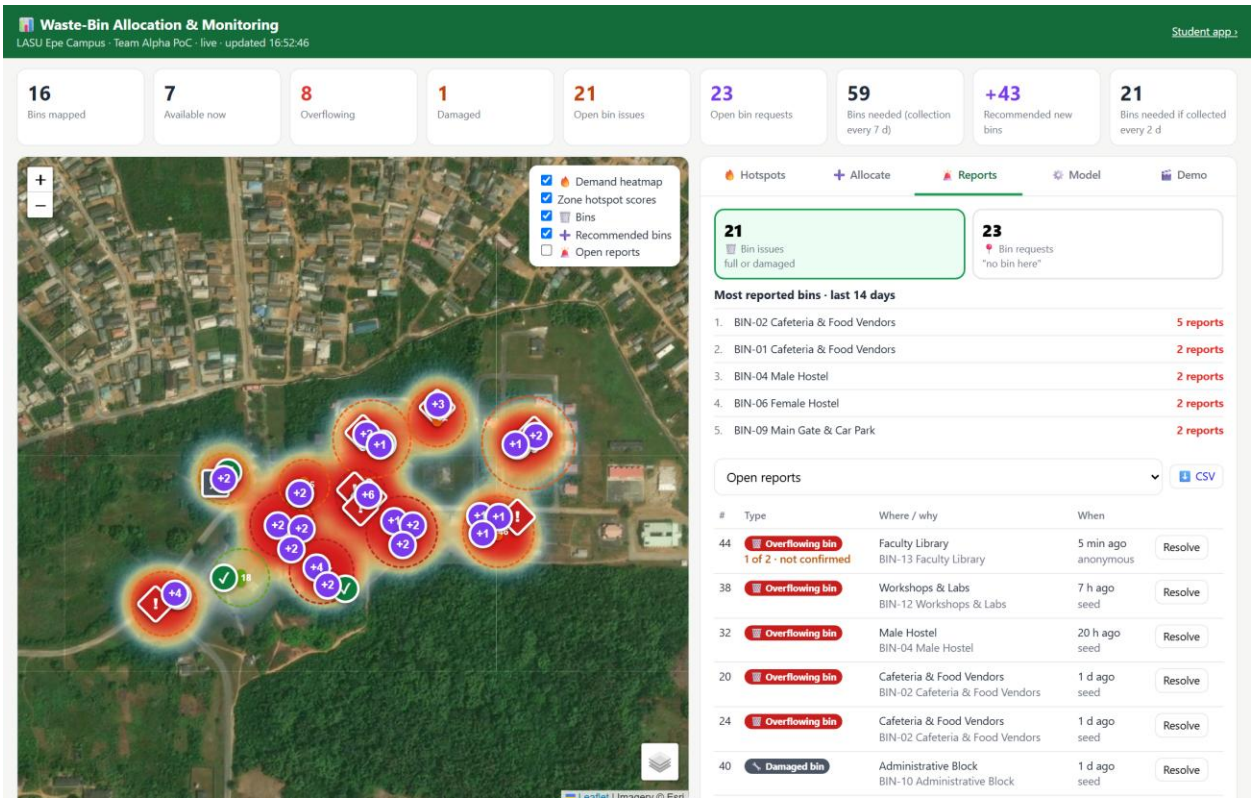
11. One report is not enough: the student is told it is counted and what happens next. (Section 4)



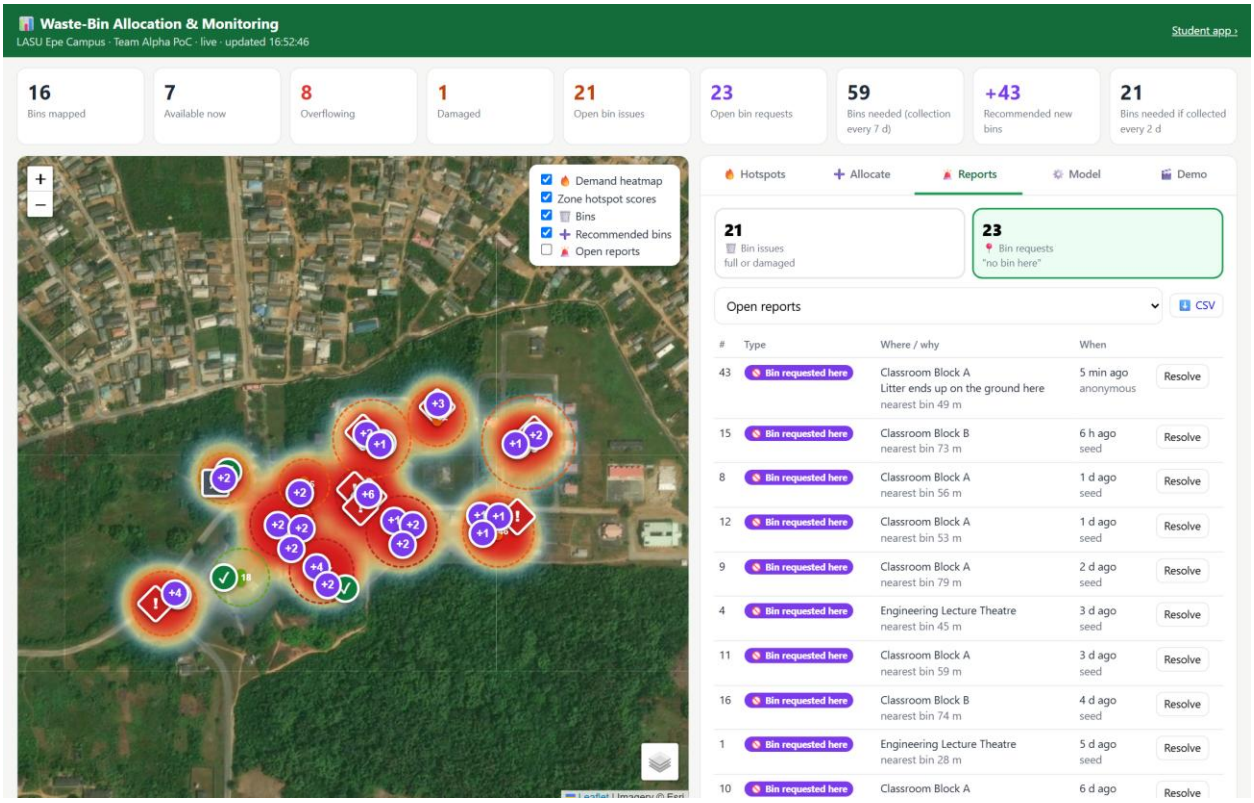
12. The bin stays available, wears an amber ring, and says why. (Section 4)

6. The updated admin dashboard

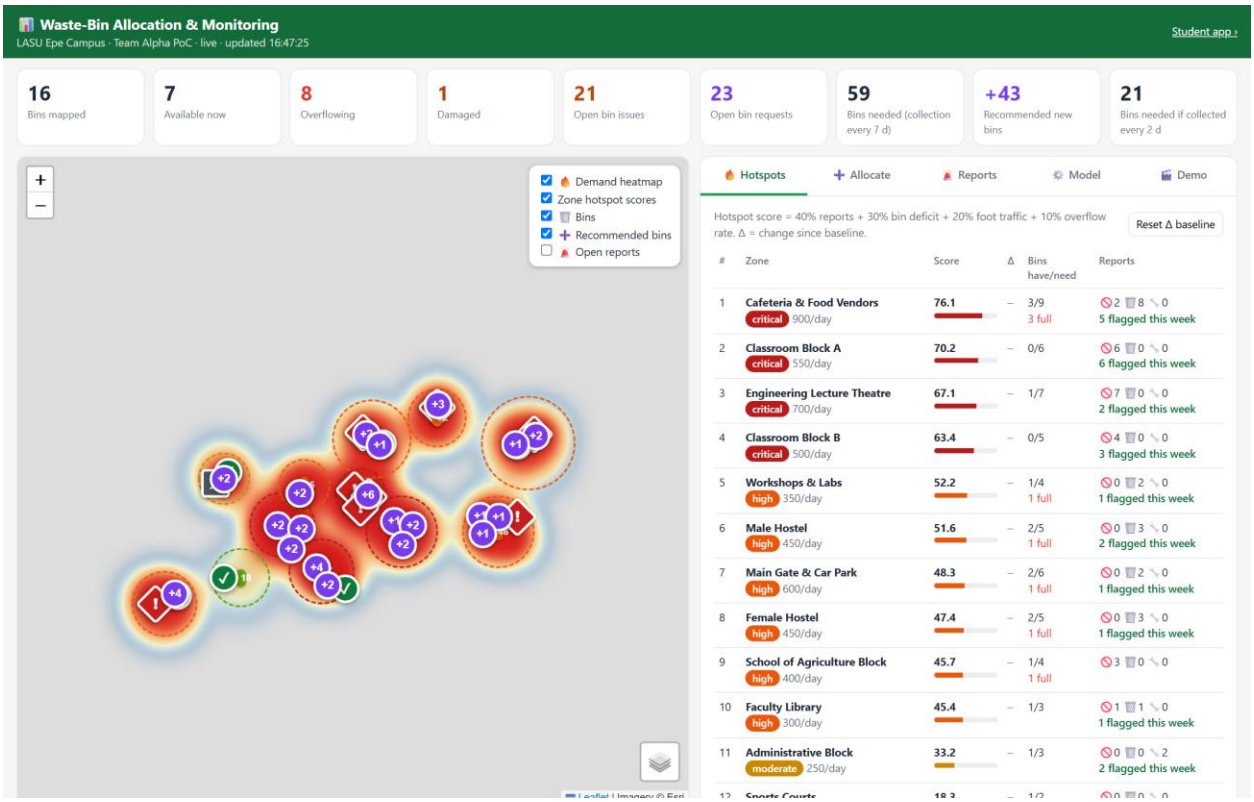
The app is not in students' hands yet, so the figures below come from the 14-day demo data set the project ships with — the rows marked “seed” are generated, not real student reports. The behaviour is live: the two streams, the rankings and the weekly counts are produced by the same code that will run on real reports.



13. Bin issues stream, with the new “Most reported bins · last 14 days” ranking above it. The KPI row counts open bin issues and open bin requests separately. (Fix 4, step 5)



14. Bin requests stream: the same table switched to “no bin here” reports, each with its reason and how far the nearest bin was. (Fix 4)



15. Hotspots, with “N flagged this week” on each zone — the line that connects student reports to the team's decisions. (Point 6)

7. What we left out, and why

Not done	Why
Turn-by-turn directions inside the app	Routing along real footpaths needs a routing service and a path network that does not exist for this campus in OpenStreetMap. The Directions button draws the straight route and distance in the app, then hands the walk to the phone's own map app. Distance is shown as 1.3 × straight-line, the standard planning allowance for a walk that is not a straight line.
Photo upload with a report	It would make reports much harder to dispute, but it needs file storage, a size limit and a moderation answer for what students might upload. Out of scope for this round; worth doing before any real deployment.
Pins reflecting a fill level rather than a status	Three statuses are what a student can judge by looking. A fill percentage would need sensors.
Phone-number or matric verification of reporters	The feedback asked for a limit, not an account system, and one report per device per bin per day handles casual distortion. It does not stop someone who clears their browser data or uses several phones; that limit is stated on the form rather than hidden.
Rebuilding the slide deck and the earlier project summary	Both still show the old three-tab screens. They will be rebuilt from these screenshots before the next presentation.

8. Two things we want to be straight about

- Two reports from different phones is a judgement call, not a measured figure. On a quiet corner of campus a real overflow might only ever be seen by one student, and that bin will keep showing as available until someone else passes. We chose that over the opposite error — one wrong report sending everyone on a

longer walk — and the number is editable, so we will tune it once we can see how often students actually report.

- The bin and zone positions are still the ones we placed from satellite imagery, not walked and recorded on the ground. Every distance the app shows is only as good as those positions. Ground-truthing them is the first thing we would do with campus access.

9. Deliverables checklist

All of it is live now at the link on the first page — it can be opened on a phone during the review.

Required	Where it is
Redesigned home screen showing the nearest available bin and its distance (Fix 1)	Screen 1
New bin-pin design showing status through colour and shape/icon (Fix 2)	Screens 1, 2, 4
Simplified bottom panel (Fix 3)	Screen 3
Nearest-bin result screen	Screen 1 — it is now the home screen
“Report a bin issue” screen	Screens 5, 6
“Request a bin here” screen	Screens 7, 8
Admin dashboard showing both report types	Screens 13, 14
Short change log: what changed, what was left out, with reasons	Sections 1–4, 7, 8